

TEMPERATURE SCALE

$$\frac{C}{5} = \frac{F-32}{9} = \frac{K-273}{5} \quad (\text{celcius-fahrenheit-kelvin conversion})$$

any scale conversion formula
Reading on any scale - lower fixed point
Upper fixed point - lower fixed point = constant

LINEAR THERMAL EXPANSION

- $\Delta l = \alpha l \Delta \theta$
- $l' = l(1 + \alpha \Delta \theta)$
- $\alpha = \frac{\Delta l}{l \Delta \theta} \rightarrow \text{unit} \rightarrow /^{\circ}\text{C or } /^{\circ}\text{K, dimension} - \text{K}^{-1}$

Whatever be the change in temperature, if the difference in length remains constant, then

$$l_1 \alpha_1 = l_2 \alpha_2$$

APPLICATIONS OF LINEAR EXPANSION

Pendulum clock
Fact $\rightarrow$ When temperature increases, time period increases, clock runs slow
 $\rightarrow$ When temperature decreases, time period decreases, clock runs fast

- Loss of time in any given time interval t ,
 $\Delta t = \frac{1}{2} \alpha \Delta \theta t$

- Time lost by clock in a day

$$\Delta t = \frac{1}{2} \alpha \Delta \theta t = \frac{1}{2} \alpha \Delta \theta 86400 = 43200 \alpha \Delta \theta$$

Thermal Stress in a rigidly fixed rod

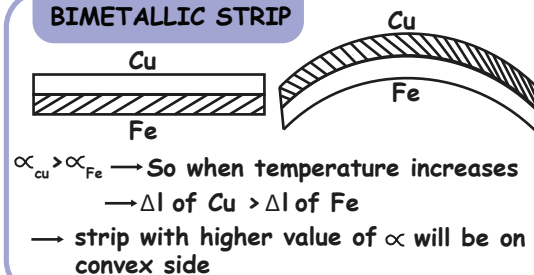
No stress in rod $\rightarrow$ Thermal stress in rod
Thermal Stress $= Y \alpha \Delta \theta$
Thermal Force $= Y A \alpha \Delta \theta$
Y - Young's Modulus
 α - coefficient of linear expansion
 $\Delta \theta$ - temperature change
A - Area of rod

ERROR IN SCALE READING DUE TO EXPANSION OR CONTRACTION

Result: At $\theta' > \theta$ True value $>$ Scale reading
At $\theta' < \theta$ True value $<$ Scale reading

$$\text{True value} = \text{Scale reading} (1 + \alpha \Delta \theta)$$

BIMETALLIC STRIP



$\alpha_{Cu} > \alpha_{Fe} \rightarrow$ So when temperature increases
 $\rightarrow \Delta l \text{ of Cu} > \Delta l \text{ of Fe}$
 $\rightarrow$ strip with higher value of α will be on convex side

EXPANSION OF CAVITY

Area of hole increases. Body expands on heating. Expansion of area of body is independent of shape and size of hole

SUPERFICIAL/AREAL EXPANSION

- $\Delta A = A \beta \Delta \theta$
- $A' = A(1 + \beta \Delta \theta)$
- $\beta = \frac{\Delta A}{A \Delta \theta} \rightarrow \text{unit} \rightarrow /^{\circ}\text{C or } /^{\circ}\text{K, dimension} - [\text{K}^{-1}]$
- $\beta = 2\alpha$

CUBICAL EXPANSION/VOLUME EXPANSION

- $\Delta V = V \gamma \Delta \theta$
- $V' = V(1 + \gamma \Delta \theta)$
- $\gamma = \frac{\Delta V}{V \Delta \theta} \rightarrow \text{unit} \rightarrow /^{\circ}\text{C or } /^{\circ}\text{K, dimension} - [\text{K}^{-1}]$
- $\gamma = 3\alpha$

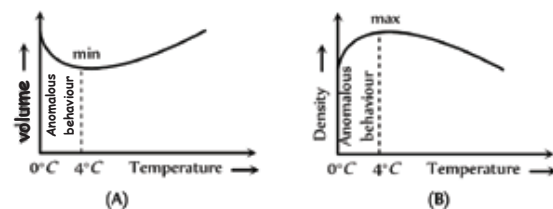
$$\alpha : \beta : \gamma = 1 : 2 : 3$$

Variation of density with temperature

Density $\propto \frac{1}{\text{Volume}}$
 $V' = V(1 + \gamma \Delta \theta)$
then $\rho' = \rho(1 - \gamma \Delta \theta)$

ANOMALOUS EXPANSION OF WATER

- Water has maximum density at 4°C (minimum volume)
- On heating, $0^{\circ}\text{C} \rightarrow 4^{\circ}\text{C}$, water contracts
 $4^{\circ}\text{C} \rightarrow$ above, water expands



REAL AND APPARENT EXPANSION OF LIQUID

- Apparent expansion of liquid $\rightarrow$ (Real expansion of liquid - expansion of solid in which liquid is contained)
- Apparent change in volume

$$\Delta V_{\text{apparent}} = V_0 \gamma_{\text{apparent}} \Delta \theta$$

$$\Rightarrow \Delta V_{\text{apparent}} = V_0 (\gamma_l - \gamma_s) \Delta \theta$$

$$\Rightarrow \Delta V_{\text{apparent}} = V_0 (\gamma_l - 3\alpha_s) \Delta \theta$$

$$\Rightarrow \gamma_{\text{apparent}} = \gamma_l - 3\alpha_s$$

γ_l - Real expansion of liquid

α_s - coefficient of linear expansion of solid

CALORIMETRY

$$1 \text{ calorie} = 4.2 \text{ J}$$

Heat Supplied (ΔQ)

change in temperature of body

- $\Delta Q = ms \Delta T$
s - specific heat capacity
SI unit - $\frac{\text{Joule}}{\text{kg Kelvin}} \rightarrow \text{J kg}^{-1} \text{K}^{-1}$
- $s_{\text{water}} = 1 \frac{\text{cal}}{\text{g}^{\circ}\text{C}} = 4.2 \frac{\text{J}}{\text{g}^{\circ}\text{C}} = 4200 \frac{\text{J}}{\text{kg}^{\circ}\text{C}}$
 $s_{\text{ice}} = \frac{1}{2} \frac{\text{cal}}{\text{g}^{\circ}\text{C}} = 2.1 \frac{\text{J}}{\text{g}^{\circ}\text{C}} = 2100 \frac{\text{J}}{\text{kg}^{\circ}\text{C}}$

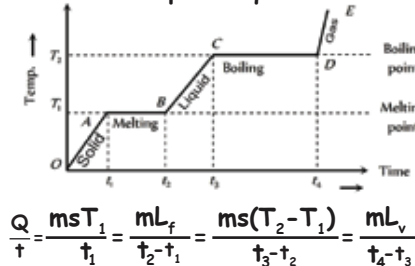
change of state of body

Melting
 $\Delta Q = mL_f$
 L_f - Latent heat of fusion

Boiling
 $\Delta Q = mL_v$
 L_v - Latent heat of vaporization

- $L_f = L_{\text{ice}} = 80 \frac{\text{cal}}{\text{g}} = 80 \times 4.2 \frac{\text{J}}{\text{g}} = 80 \times 4200 \frac{\text{J}}{\text{kg}}$
- $L_v = L_{\text{steam}} = 540 \frac{\text{cal}}{\text{g}} = 540 \times 4.2 \frac{\text{J}}{\text{g}} = 540 \times 4200 \frac{\text{J}}{\text{kg}}$

Heat supplied at constant rate
Graph & equation



$$\frac{Q}{t} = \frac{msT_1}{t_1} = \frac{mL_f}{t_2 - t_1} = \frac{ms(T_2 - T_1)}{t_3 - t_2} = \frac{mL_v}{t_4 - t_3}$$

if specific heat is variable $\Delta Q = \int_{T_1}^{T_2} msdT$

HEAT CAPACITY

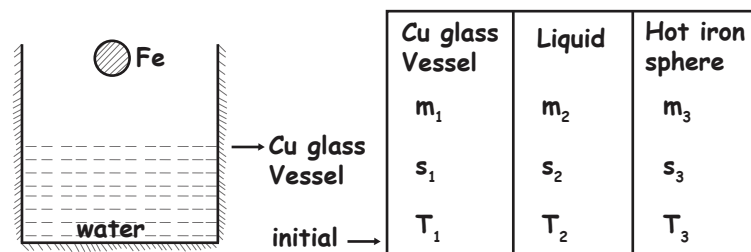
Heat capacity = mass $\times$ specific heat capacity
Unit = $\frac{\text{cal}}{^{\circ}\text{C}}$, SI unit is $\frac{\text{J}}{\text{K}}$

WATER EQUIVALENT

The mass of water that will absorb or lose the same quantity of heat as a given substance will do for same change in temperature

$$m_w s_w = m_b s_b \quad \begin{matrix} w = \text{water} \\ b = \text{body} \end{matrix}$$

PRINCIPLE OF CALORIMETRY



Heat lost by the hotter bodies = Heat gained by colder bodies
 $Q_3 = Q_1 + Q_2$

Final equilibrium temperature,

$$T_{\text{eq}} = \frac{m_1 s_1 T_1 + m_2 s_2 T_2 + m_3 s_3 T_3}{m_1 s_1 + m_2 s_2 + m_3 s_3} = \frac{\sum msT}{\sum ms}$$

Facts:
Calorimeter - A device for measurement of amount of heat involved in a process.

ICE-WATER SYSTEM

Problem solving methodology

- m_1 gram ice [$- \theta_1^{\circ}\text{C}$] mixed with m_2 gram water [$\theta_2^{\circ}\text{C}$]

- Convert $- \theta_1^{\circ}\text{C}$ ice $\rightarrow 0^{\circ}\text{C}$ ice

$$\Delta Q_1 = m_1 s_{\text{ice}} \theta_1$$

- Convert 0°C ice $\rightarrow 0^{\circ}\text{C}$ water

$$\Delta Q_2 = m_1 L_f$$

- Convert $\theta_2^{\circ}\text{C}$ water $\rightarrow 0^{\circ}\text{C}$ water

$$\Delta Q_3 = m_2 s_{\text{water}} \theta_2$$

check $\Delta Q_3 > \Delta Q_1 + \Delta Q_2$ or $\Delta Q_1 + \Delta Q_2$

$$\Delta Q_3 > \Delta Q_1 + \Delta Q_2$$

- Whole ice melts into water
- Additional heat [$\Delta Q' = \Delta Q_3 - (\Delta Q_1 + \Delta Q_2)$] is used to increase the temperature of system from 0°C
- Final temperature can be found out by
 $\Delta Q' = M_{\text{total}} s_{\text{water}} T$

$$\Delta Q_3 < \Delta Q_1 + \Delta Q_2$$

- Only m' g of ice melts
- Mass of ice melted can be found by
 $[m = \text{mass of ice melted}]$
 $m L_f = Q$
- Final temperature is 0°C

CONVERSION OF MECHANICAL ENERGY TO HEAT ENERGY

- Potential energy to heat energy
 $\Delta U = mgh \xrightarrow{\text{converts to heat}} \Delta Q = m' L_f$ [m' = mass of substance melted/vaporized]
When equating, multiply ΔQ with 4200 J if L_f is in cal/g
i.e., $mgh = m' L_f \times 4200$
- Kinetic energy to Heat energy
 $K.E = \frac{1}{2} mv^2 \xrightarrow{\text{converts to heat}} \Delta Q = m' L_f$ [m' = mass of substance melted/vaporized]
If L_f is in $\frac{\text{calorie}}{\text{g}}$
then $\frac{1}{2} mv^2 = m' L_f \times 4200$

HEAT TRANSFER

- Conduction:
Heat flows from hot end to cold end.
Medium is necessary.
Slow process.

$$\frac{dQ}{dt} = K A \frac{d\theta}{dx}$$

Unit of 'K' = $\frac{\text{watt}}{\text{metre}^{\circ}\text{C}}$ or $\frac{\text{watt}}{\text{metre K}}$

'K' depends on the nature of material

$\frac{dQ}{dt}$ = Rate of flow of heat

A = Area of cross section

$\frac{d\theta}{dx}$ = Temperature gradient

K = coefficient of thermal conductivity

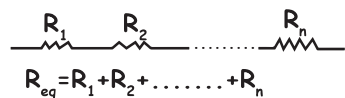
THERMAL PROPERTIES OF MATTER

OHM'S LAW OF CONDUCTION

Electrical Conduction

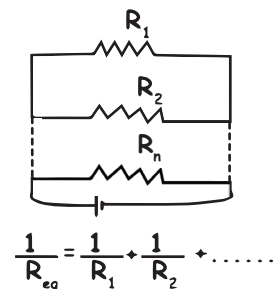
- 1) current, $I = \frac{dq}{dt}$
- 2) $I = \frac{\Delta V}{R}$ ($\Delta V = V_{\text{high}} - V_{\text{low}}$)
- 3) electrical resistance, $R = \frac{\rho l}{A}$
- 4) $I = \frac{V_1 - V_2}{R} = \frac{(V_1 - V_2)A}{\rho l} = \frac{\sigma A}{l} (V_1 - V_2)$
- 5) Combination of resistors

i) Series Combination



Here 'I' is same in all resistors

ii) Parallel Combination

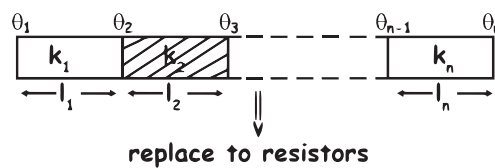


Here $(V_1 - V_2)$ is same for all resistors

Thermal Conduction

- 1) Heat current, $H = \frac{dQ}{dt}$
- 2) $H = \frac{\theta_1 - \theta_2}{R} = \frac{\Delta \theta}{R}$ ($\theta_1 > \theta_2$)
- 3) Thermal resistance, $R = \frac{l}{KA}$
- 4) $H = \frac{\theta_1 - \theta_2}{R} = \frac{\theta_1 - \theta_2}{(l/K_A)} = \frac{KA}{l} (\theta_1 - \theta_2)$
- 5) Combination of conductors

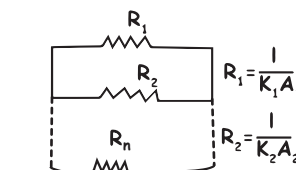
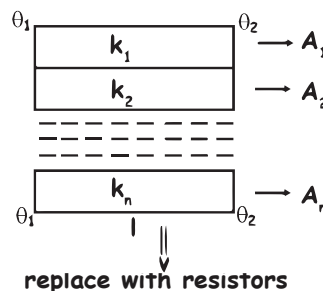
i) Series Combination



$$R_1 = \frac{l_1}{k_1 A} \quad R_2 = \frac{l_2}{k_2 A} \quad \dots$$

Find $R_{eq} = R_1 + R_2 + \dots$ Here, heat current, H is same in all conductors

ii) Parallel Combination



$$\text{Find } \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

from that find K_{eq}

Here, Temp Difference is same for all conductors

CONVECTION

Requires a medium. Actual movement of fluid. Occurs naturally or forced.

Natural convection takes place due to the effect of gravity

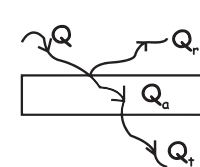
Applications:

Sea Breeze
Wind blows from sea to land during day time

Land Breeze
Wind blows from land to sea during night

RADIATION

Absorptive, reflective and Transmittive power



$$a + r + t = 1$$

$$\text{Absorptive power}(a) = \frac{Q_a}{Q} = \frac{\text{Energy absorbed}}{\text{Energy incident}}$$

$$\text{Reflective power}(r) = \frac{Q_r}{Q} = \frac{\text{Energy reflected}}{\text{Energy incident}}$$

$$\text{Transmittive power}(t) = \frac{Q_t}{Q} = \frac{\text{Energy transmitted}}{\text{Energy incident}}$$

EMISSIVE POWER/INTENSITY OF THERMAL RADIATION

$$\text{Emissive power}(E) = \frac{\text{Energy radiated}}{\text{area} \times \text{time}} \quad \text{unit} \rightarrow \frac{\text{Watt}}{\text{m}^2}$$

$$\text{Spectral emissive power}(E_\lambda) = \frac{\text{Energy radiated}}{\text{area} \times \text{time} \times \text{wavelength}} \quad \text{unit} \rightarrow \frac{\text{Watt}}{\text{m}^3}$$

$$\text{Relation between } E \text{ \& } E_\lambda \implies E = \int_0^\infty E_\lambda d\lambda$$

EMISSIVITY (e)

$$e = \frac{\text{Energy radiated by a general body}}{\text{Energy radiated by a black body}}$$

value of $e \implies 0 < e < 1$

If $e = 0$, the body radiates no energy

If $e = 1$, the body is a perfect black body

KIRCHHOFF'S LAW

Ratio of emissive power to absorptive power is same for all surfaces at the same temperature and is equal to the emissive power of a perfectly black body at that temperature.

$$\frac{E_1}{a_1} = \frac{E_2}{a_2} = \dots = E_b$$

STEFAN'S LAW

Emissive power of a black body $\propto$ fourth power of absolute temperature and surface area of the body

$$E = \sigma AT^4 \quad \text{OR} \quad \frac{\Delta Q}{\Delta t} = \sigma AT^4$$

$$\sigma \rightarrow \text{Stefan's constant} \quad \frac{\Delta Q}{\Delta t} \rightarrow \text{Radiant power}$$

$$\text{value of } \sigma \rightarrow 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

$$\text{Dimension} \rightarrow [\sigma] = \text{MT}^{-3} \text{K}^{-4}$$

For ordinary body $E = e\sigma T^4$

$$\frac{\Delta Q}{\Delta t} = eA\sigma T^4 \quad e = \text{emissivity}$$

In the presence of a surrounding. (T_0 = Surrounding temperature)
For black body,

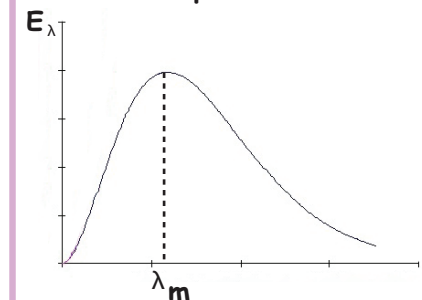
$$E = \sigma A(T^4 - T_0^4)$$

In the presence of a surrounding. (T_0 = Surrounding temperature)
For general body,

$$E = e\sigma A(T^4 - T_0^4)$$

WIEN'S LAW

Wien's displacement law



$$\lambda_m \propto \frac{1}{T}$$

$$\lambda_m T = b$$

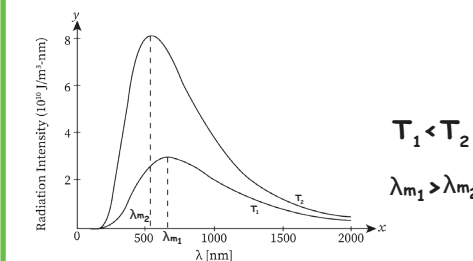
b = Wien's constant

$$\text{Hence } \frac{A_1}{A_2} = \left[\frac{T_1}{T_2} \right]^4$$

$$b = 2.89 \times 10^{-3} \text{ mK}$$

$$\lambda_m T_1 = \lambda_m T_2 \quad \text{Area under the graph, } A = \int_0^\infty E_\lambda d\lambda = E = \sigma T^4 \quad [\text{Dimensions}] = [b] = [\text{LK}]$$

"As the temperature of the body increases, the wavelength at which the spectral intensity (E_λ) is maximum shift towards left."



NEWTON'S LAW OF COOLING

Rate of cooling $\propto$ excess temperature of the body over the surrounding.

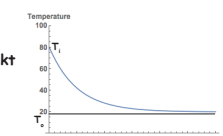
$$\frac{-dT}{dt} \propto (T - T_0)$$

T = Temperature of body

T_0 = Temperature of surrounding

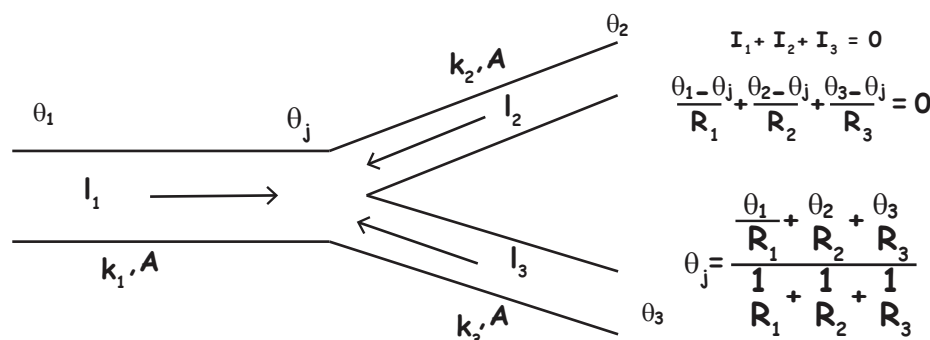
T_i = initial temperature of the body

$$\frac{T - T_0}{T_i - T_0} = e^{-kt}$$



THERMAL PROPERTIES OF MATTER

TEMPERATURE OF INTERMEDIATE JUNCTION



$$I_1 + I_2 + I_3 = 0$$

$$\frac{\theta_1 - \theta_j}{R_1} + \frac{\theta_2 - \theta_j}{R_2} + \frac{\theta_3 - \theta_j}{R_3} = 0$$

$$\theta_j = \frac{\frac{\theta_1}{R_1} + \frac{\theta_2}{R_2} + \frac{\theta_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$